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Information Security

Assignment #1

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Instructor Name:

(Q=1)

a) Violated Principles: Confidentiality is violated because patient records were copied without permission. Alteration of prescriptions lead to integrity violation. Availability is violated as the server was down.

b) Cryptographic Mechanisms:

Use AES encryption for confidentiality, SHA-256 or HMAC for integrity, and digital signatures for authenticity and authorization control.

c) Key Management Impact: If keys are not rotated or protected properly, attackers can decrypt data. Strong key storage and periodic key change make encryption effective.

(Q=2)

a) CIA Impact: Confidentiality may be breached if data is stolen. Integrity affected because files are modified and availability is heavily affected since systems are locked.

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b) Layered Strategy: Use encryption for stored data, maintain offline hashed backups, and apply multi-factor authentication. Add Firewalls and intrusion detection.

c) Strategy vs Policy:

Security Strategy	Strategy Policy
1) Practical Implementation of controls.	Written Rules and guidelines
2) Technical & operational	Administrative & formal
3) Example: Backup + MFA	Example: Password Policy.

← (Q=3) →

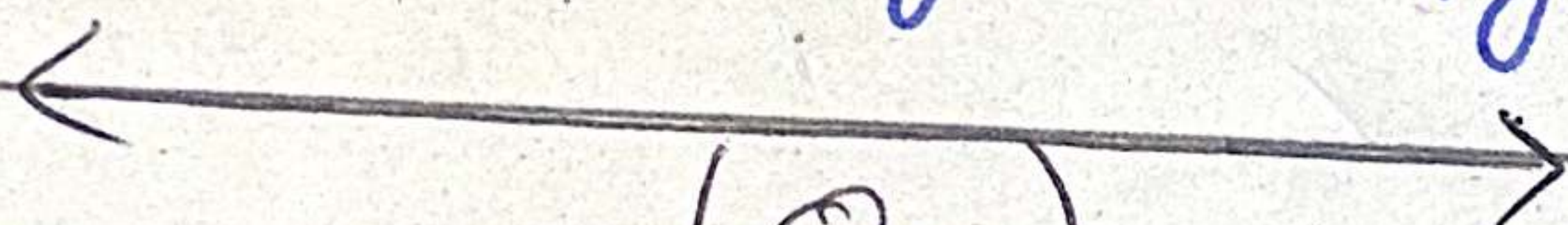
Cyber Security	Information Security
1) Protect digital systems	Protect all information.
2) Focus on networks and cyber threats.	Cover physical, admin and cyber risks.
3) Example: Online banking firewall.	Example: Locked-file cabinets.

Information Security is broader because it includes both digital & non-digital protection. Both aim to protect CIA principles.

(Q=4)

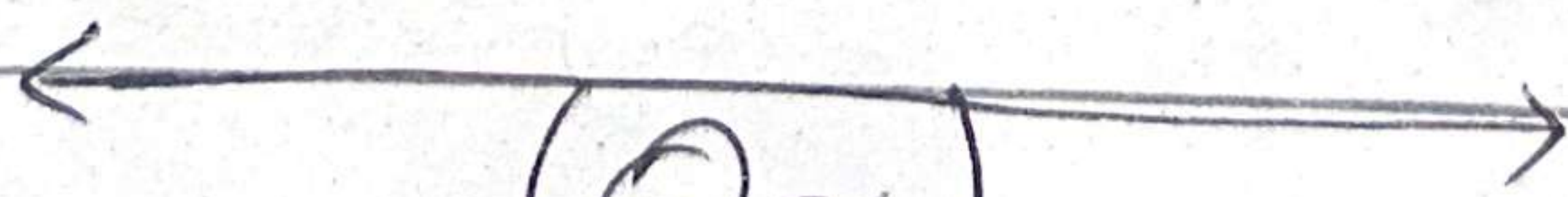
IOT without Encryption.

- a) Challenges: IOT devices have weak processing power, poor updates and weak authentication, making them easy targets.
- b) Confidentiality Risks: If plaintext P is sent, attacker reads it directly since $C=P$. Without Encryption, secrecy is zero.
- c) Encryption Choice: Use symmetric AES for fast data encryption and asymmetric RSA only for secure-key exchange.



(Q=5)

Confidentiality keeps vote secret, integrity prevents vote tampering. Availability ensures system uptime, authenticity, Integrity is hardest because even small changes affect results. Use AES for secrecy, hashing for integrity, digital signatures for authenticity, PKI for authorization and redundancy for availability.



(Q=6)

Reusing one key for 10 years increases risks of exposure, so it is poor key management.

AES-128 Key space = $2^{128} \approx 3.4 \times 10^{38}$ keys.

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Brute force is impractical, but long-term reuse increases compromise chances.

$(Q=7)$	
Symmetric	Asymmetric
1) Faster & efficient	Slower & complex
2) Very fast	Slower
3) Used for Bulk Data	Used for key exchange

TLS uses hybrid encryption where RSA exchanges key and AES encrypts data.

DES Security:-

$$\text{Total Keys} = 2^{56} \approx 7.2 \times 10^{16}$$

DES is insecure because modern systems can brute-force it. AES-256 with 2^{256} keys is exponentially stronger.

$$\begin{aligned} & \leftarrow (Q=9) \rightarrow \\ n &= 11 \times 19 = 209 \\ \phi(n) &= 10 \times 18 = 180 \\ e=7 & \text{ is valid since } \gcd(7, 180) = 1 \\ d &= 103 \end{aligned}$$

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Encrypted $M=9 \rightarrow C=48$, Decrypted $\rightarrow 9$

$(Q=10)$

Signature: $S = H(M)^d \bmod n$

Verification: $H(M) = S^e \bmod n$

Provides non-repudiation because only private key holder can sign.

$(Q=11)$

Hashing is one-way because original message cannot be derived from the hash.

~~SHA-256~~ SHA-256 is collision resistant unlike MD5. Collision probability follows Birthday paradox $\approx \sqrt{2^{256}}$

$(Q=12)$

MD5 is vulnerable to collision attacks where two messages produce same hash.

It is also weak against rainbow table.

Use bcrypt or SHA-256 with salt instead

$(Q=13)$

Formula

$$HMAC(K, M) = H((K \oplus \text{opad}) \parallel H((K \oplus \text{ipad}) \parallel M))$$

HMAC is more stronger and secure than the simple MAC because it resists length extension attacks.

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(Q=14)

PGP Encryption:-

PGP uses hybrid encryption. RSA encrypts session key, AES encrypts message. Public key encrypts, private key decrypts. Trust model is web of trust instead of centralized authority.

← (Q=15) →

AES Brute Force:

$$\text{Time} = 2^{128} / 10^{12} \approx 10^{19} \text{ years}$$

Brute force tests all keys. Differential cryptanalysis analyses patterns. Mitigation includes strong keys and key rotations.

← (Q=16) →

Symmetric key distribution requires $n(n-1)/2$ keys.

$$\text{For 1000 users: } \frac{1000 \times 999}{2} = 499,500 \text{ keys.}$$

PKI solves this using public private keys and certificates.

← →

(Q=17)

DDoS Attacks:-

Availability principle is violated. Cryptographic mitigation includes authentication to filter fake requests. Non-cryptographic includes rate limiting and firewalls.

Encryption alone cannot stop traffic flooding.

← (Q=18) →

Blockchain Hashing:-

Integrity ensured because each block stores hash of previous block.

Hash chaining:

$$\text{Hash-n} = H(\text{Block-n} \parallel \text{Hash-n-1})$$

If one block changes, all subsequent hashes break.

← (Q=19) →

SubBytes substitutes bytes using S-box.

ShiftRows shifts rows.

MixColumns mixes columns mathematically.

AES is stronger due to larger key size and better design.

Key expansion generates round keys.

(Q=20)

AuthorizationAuthentication

- | | |
|--|---|
| i) Authentication verifies identity. | Authorization grants permission. |
| ii) Authentication supports authenticity | Authorization supports integrity and confidentiality. |
| iii) Ex:- Login verifies identity | Ex: Access Control define actions. |

(Q=21)

Modern Cryptography:-

Block cipher encrypts fixed blocks like ~~AE~~ AES. Stream cipher encrypts bit by bit. Public-key cryptography uses key pairs like RSA.

Symmetric is faster. Public key is slower.

(Q=22)

- 1) Broadcast attack occurs when same message encrypted with small exponent.
- 2) Mathematically, attacker solves using Chinese Remainder Theorem.
- 3) Secure padding such as OAEP prevents this.

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(Q=23)

Certificate Expiry: Certificate lifecycle includes issuance, usage, expiration, renewal, revocation. During TLS handshake, certificate is verified. Revocation via CRL & OCSP.

← (Q=24) →

Confidentiality vs Privacy:

<u>Confidentiality</u>	<u>Privacy</u>
i) Technical Protection	Legal & ethical concept.
ii) Prevents unauthorized access	Protects personal rights
iii) Achieved using encryption	supported by laws & policies.

Confidentiality protects data from unauthorized data, & privacy protects personal data. Encryption protects privacy.

← (Q=25) →

Caesar Cipher:

Frequency suggest shift of 3.

Decryption = KHHOR ZRUOG shifted back by 3 becomes HELLO WORLD.

Caesar Cipher has only 25 possible keys, making brute-force easy. More ciphers like AES use extremely large key spaces, making them secure.

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(Q=26)

SHA-256 Structure:-

- Uses comparison function iteratively.
- 256-bit output provides high collision resistance.
- Avalanche effect means small change causes large hash-difference.

← (Q=27) →

DES vs AES vs RSA:-

- DES vs AES are symmetric. RSA is asymmetric.
- DES uses 56-bit key. AES uses 128-256 bits. RSA uses 2048+ bits.
- AES is most efficient for bulk encryption.

← (Q=28) →

Shared Symmetric Key:-

$$MAC = H(K \parallel M)$$

If K leaks, attacker can forge MAC.
Forward secrecy ensures past sessions remain secure even if key leaks.

← (Q=29) →

National Bank Security Strategy:

The bank must identify threats like phishing, insider attacks, and malware. Risk analysis

helps prioritise security investments. Defence in depth uses multiple layers. Incident response includes encrypted backups and recovery plans. Encrypted data ensures data recovery after attacks.

(Q=30)

Fintech Architecture

AES encrypts data.

RSA exchanges keys

SHA-256 ensures integrity.

HMAC authenticates APIs.

If key-management fails, entire system collapses. Improvements include key-rotation, hardware security modules, and quantum-resistant algorithms. The company should implement these. Continuous monitoring and security audits improve long-term resilience.

